Sebastian Capellan

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EDUCATION

New York University New York, NY
B.A. Data Science & Computer Science, GPA: 3.55 Expected May 2028
Relevant Coursework: Data Management, Operating Systems, Systems Programming, Computer Architecture, Data Structures, Data Science, Open Source Software Development, Basic Algorithms, Multivariable Calculus, Discrete Math, Linear Algebra

WORK EXPERIENCE

Duolingo Pittsburgh, PA
Incoming Software Engineer Intern June 2026
University of California, San Francisco Remote
Research Assistant October 2025 - April 2026

- Migrated a Next.js microplastics research dashboard from per-request JSON/CSV file parsing to an indexed SQLite database via custom API routes and seeding, reducing API response times by 70+%.
- Engineered a universal Python ETL pipeline that auto-infers schemas from heterogeneous JSON sources, normalizes nested objects and arrays into relational SQLite tables, and validates row counts post-migration across 10+ tables.
- Fact-checked and corrected physical property data across 18 polymer records against peer-reviewed literature and manufacturer datasheets.

PROJECTS

Multithreaded Key-Value Server | C, POSIX, Ubuntu Linux

- Designed and implemented a datastore in C, supporting read-write functionality via CLI parsing and signaling.
- Engineered core functionality using POSIX system calls, enabling interaction with the OS for file persistence, concurrency, and IPC.
- Implemented datastore networking, supporting multiple clients through UNIX domain sockets and multithreaded concurrency.

Interactive Coding Practice Platform | Python, Flask, HTML, CSS

- Developed a web-based, CLI style application, that parses user submitted LeetCode JSON files in order to retrieve random questions for review from the site, allowing active recall for website users.
- Launched publicly and achieved 150+ unique users, demonstrating adoption, usability, and real world impact.

Cryptocurrency Portfolio Tracker | Java, Spring, MySQL

- Engineered "Coinleaf," a backend cryptocurrency portfolio tracker with RESTful API endpoints using Java, Spring Boot, and MySQL, enabling full CRUD operations and database persistence for 100% of portfolio data.
- Integrated live market data by consuming Coinbase API services, building a service layer to fetch and update real-time cryptocurrency prices with <200ms response times, improving accuracy of portfolio valuation.

Mandelbrot Fractal Visualizer | Python, Streamlit, MatPlotLib, NumPy, Numba

- Developed and optimized fractal generation algorithm using Numba JIT compilation, NumPy vectorization, and dynamic iteration control based on zoom depth, achieving over 700% image rendering speedup.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

ColorStack @ NYU New York, NY
President & Co-Founder January 2025 - Present

- Foster an NYU specific chapter of an organization with 16,000+ members nationally, providing peer support, career development, and technical skill-building for the empowerment of Black and Latino students in tech.
- Spearhead outreach activity, gathering 60+ student members through targeted engagement and event hosting.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C, Java, HTML, CSS
Technologies: MySQL, Linux, Docker, AWS, Git, GitHub
Libraries & Frameworks: Spring, Flask, NumPy, Pandas